

```
In [1]: import nltk
```

```
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('averaged_perceptron_tagger')
```

```
[nltk_data] Downloading package punkt to
[nltk_data]   C:\Users\System05\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data]   C:\Users\System05\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data]   C:\Users\System05\AppData\Roaming\nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]   C:\Users\System05\AppData\Roaming\nltk_data...
[nltk_data]   Package averaged_perceptron_tagger is already up-to-
[nltk_data]   date!
```

```
Out[1]: True
```

```
In [4]: text = "Tokenization is the first step in text analytics.The process of breaking do
```

```
In [5]: import nltk
```

```
nltk.download('punkt')
nltk.download('punkt_tab')
```

```
[nltk_data] Downloading package punkt to
[nltk_data]   C:\Users\System05\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to
[nltk_data]   C:\Users\System05\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
```

```
Out[5]: True
```

```
In [6]: from nltk.tokenize import sent_tokenize
```

```
sentences = sent_tokenize(text)
print(sentences)
```

```
['Tokenization is the first step in text analytics.The process of breaking down a text paragraph into smallerchunks such as words or sentences is called Tokenization.']
```

```
In [7]: from nltk.tokenize import word_tokenize
```

```
words = word_tokenize(text)
print(words)
```

```
['Tokenization', 'is', 'the', 'first', 'step', 'in', 'text', 'analytics.The', 'proces', 's', 'of', 'breaking', 'down', 'a', 'text', 'paragraph', 'into', 'smallerchunks', 's', 'uch', 'as', 'words', 'or', 'sentences', 'is', 'called', 'Tokenization', '.']
```

```
In [9]: import re
        from nltk.corpus import stopwords
        stop_words = set(stopwords.words("english"))
        text_clean = re.sub('[^a-zA-Z]', ' ', text)
        tokens = word_tokenize(text_clean.lower())
        filtered = []
        for w in tokens:
            if w not in stop_words:
                filtered.append(w)
        print(filtered)
```

```
['tokenization', 'first', 'step', 'text', 'analytics', 'process', 'breaking', 'text', 'paragraph', 'smallerchunks', 'words', 'sentences', 'called', 'tokenization']
```

```
In [10]: from nltk.stem import PorterStemmer

        ps = PorterStemmer()

        for w in filtered:
            print(ps.stem(w))
```

```
token
first
step
text
analyt
process
break
text
paragraph
smallerchunk
word
sentenc
call
token
```

```
In [11]: from nltk.stem import WordNetLemmatizer

        lemma = WordNetLemmatizer()

        for w in filtered:
            print(lemma.lemmatize(w))
```

```
tokenization
first
step
text
analytics
process
breaking
text
paragraph
smallerchunks
word
sentence
called
tokenization
```

In [12]: `import nltk`

```
nltk.download('averaged_perceptron_tagger')
nltk.download('averaged_perceptron_tagger_eng')
```

```
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data] C:\Users\System05\AppData\Roaming\nltk_data...
[nltk_data] Package averaged_perceptron_tagger is already up-to-
[nltk_data] date!
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data] C:\Users\System05\AppData\Roaming\nltk_data...
[nltk_data] Package averaged_perceptron_tagger_eng is already up-to-
[nltk_data] date!
```

Out[12]: `True`

In [13]: `import nltk`

```
pos = nltk.pos_tag(words)

print(pos)
```

```
[('Tokenization', 'NN'), ('is', 'VBZ'), ('the', 'DT'), ('first', 'JJ'), ('step', 'N'), ('in', 'IN'), ('text', 'NN'), ('analytics.The', 'NN'), ('process', 'NN'), ('of', 'IN'), ('breaking', 'VBG'), ('down', 'RP'), ('a', 'DT'), ('text', 'NN'), ('paragraph', 'NN'), ('into', 'IN'), ('smallerchunks', 'NNS'), ('such', 'JJ'), ('as', 'IN'), ('words', 'NNS'), ('or', 'CC'), ('sentences', 'NNS'), ('is', 'VBZ'), ('called', 'VBN'), ('Tokenization', 'NN'), ('.', '.')]

```

In []: